

PROJECT PROPOSAL

Cinema Management & Integrated Booking System

Group Members

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1. System Overview

The envisioned system is a centralized platform designed to streamline cinema operations while providing a seamless user experience for customers. The system manages the full lifecycle of a movie—from its theater run to daily screenings—and facilitates the entire booking process, including seat selection and concession sales. Administratively, it handles organizational needs such as staff authorization (via a hierarchical approval system), inventory management for retail items, and theater-specific resource allocation.

2. Core Functionalities

Hierarchical User Management: Utilizing an ISA relationship, the system distinguishes between Customers (with loyalty tracking and favorites) and Employees. Staff accounts are governed by an `auth_level` (Authority Level) and `account_status` (Pending/Active) to ensure secure, role-based access control.

Physical Asset Mapping: Using Weak Entities, the database precisely maps Theaters to their Saloons and individual Seats. This ensures physical constraints (e.g., specific seating layouts) are maintained and prevents double-booking.

Dynamic Scheduling: The system tracks Movie Runs (active date ranges) and individual Screenings, allowing for price variations based on movie formats (IMAX, 3D, 2D) and subtitling options.

Integrated POS & Retail: The Booking hub manages the sale of Tickets and Consumables (concessions). The system automatically tracks stock quantities and applies promotional Deals to total amounts.

Feedback & Metadata: Customers can submit Reviews, while movie data (cast, summaries, directors) is envisioning to be automatically synchronized with IMDB to maintain a professional catalog.

3. Database Design Highlights

- **Specialization/Generalization:** A User superclass ensures consistent authentication across all roles.
- **Weak Entities:** Saloon, Seat, Ticket, and Payment entities maintain strict relational dependency for data integrity.
- **Complex Relationships:** A Many-to-Many (M:N) favorites relationship for customers and a recursive approves relationship for admin-led staff onboarding.
- **Integrity Constraints:** Use of NOT NULL constraints, derived attributes (e.g., total_amount), and Total Participation constraints to ensure every booking is linked to a registered user.

4. Entity & Relationship Detail

I. User & Personnel Hierarchy

User (Base Entity)

Attributes: user_id (PK), first_name, last_name, phone_number, email, password, created_at.

Employee (Subclass)

Attributes: role, salary, work_shift, auth_level, account_status.

Relation: ISA User | Recursive 'approves' relation

Customer (Subclass)

Attributes: *birth_date*, *loyalty_points*, *membership_tier*.

Relation: ISA User

II. Movie Content & Metadata

Movie (Entity)

Attributes: *movie_id* (PK), *title*, *director*, *duration_mins*, *rating_age*, *release_date*, *summary*, {*Cast*} (Multivalued).

Genre & Format (Entities)

Attributes: *id* (PK), *name*.

Relation: 'of' (Genre), 'with' (Format)

Review (Entity)

Attributes: *review_id* (PK), *rating*, *comment*.

Relation: 'submit' (Customer), 'for' (Movie)

III. Physical Infrastructure

Theater (Entity)

Attributes: *theater_id* (PK), *name*, *phone_number*, {*address*} (Multivalued).

Saloon (Weak Entity)

Attributes: *number* (Partial Key), *capacity*, *type*, *isActive*.

Identifying Relation: 'has' (Theater)

Seat (Weak Entity)

Attributes: *row_letter* (PK), *number* (PK), *type*.

Identifying Relation: 'contains' (Saloon)

IV. Scheduling & Operations

Movie_Run (Weak Entity)

Attributes: *start_date*, *end_date*.

Identifying Relation: 'plays' (Movie)

Screening (Entity)

Attributes: *screening_id* (PK), *start_time*, *base_price*, *isSubtitled*.

Relation: 'shows' (Movie), 'takes_place_in' (Saloon)

V. Transactions & Retail

Booking (Entity)

Attributes: *booking_id* (PK), *timestamp*, *total_amount* (Derived).

Relation: 'makes' (Customer - Total Participation)

Ticket & Payment (Weak Entities)

Attributes: *type* (Ticket); *method*, *status* (Payment).

Identifying Relations: 'contains' / 'via' (Booking)

Consumable & Deal

Attributes: *id* (PK), *name*, *price/discount*.

Relation: 'added' to Booking (Consumable), 'applies_to' Booking (Deal)

5. Planned System Architecture (Sitemap)

Customer Portal

- Home Page
- Movie Discovery (Now Playing / Coming Soon)
- Movie Detail (Cast, IMDB Metadata, Reviews)
- Seat Selection & Ticket Type Selection
- Integrated Checkout (Payment & Concessions)
- Profile (Booking History, Refund Request, Favorites, Edit Info)

Staff Interface

- Login / Dashboard
- Manual Booking (for Walk-in Customers/Guests)
- Refund Management
- Ticket Validation (QR/Code Scan)

Admin Panel

- Account Management (Staff Approval & Hierarchy)
- Movie & Screening Management (Edit/Add Movie, Schedule Shifts)
- Saloon & Infrastructure Setup
- Inventory Management (Consumables & Stock)
- Promotional Deal Configuration